Exit Ticket Answers

What are the hallmarks of good exit ticket answers?

Characterize the features of good exit ticket answers

ChatGPT (Medium):

Good exit ticket answers should be **short, diagnostic, and explanatory**. The goal is not to see whether students can reproduce lecture notes; it is to see whether they understand the key idea well enough to express it correctly with minimal prompting.

A strong exit-ticket answer usually has these features:

- **Correct conclusion.** The answer reaches the right result: stable/unstable, aliases to 3 kHz, DTFT exists/does not exist, zero creates a notch, etc.
- **One essential reason.** Students should give the key reasoning, not just the final word. For example, "Stable because the unit circle lies in the ROC" is much more informative than simply "stable."
- **Uses the right concept or vocabulary.** Terms such as ROC, unit circle, pole, zero, aliasing, sinc, quantization, and causality should be used accurately.
- **Shows the relevant connection.** Good answers connect two ideas: "causal ⇒ exterior ROC," "zero on unit circle ⇒ numerator becomes zero," or "adding one bit halves (\Delta) ⇒ about 6 dB SQNR improvement."
- **Avoids irrelevant detail.** An exit ticket should not require a full derivation unless derivation itself is the learning objective. Usually one equation plus one sentence is enough.
- **Makes the reasoning visible.** You want enough evidence to distinguish understanding from guessing.

For your DSP lectures, I would think of answers in roughly three quality levels:

Weak answer: "Stable."

Acceptable answer: "Stable because all the poles are inside the unit circle."

Strong answer: "For a causal rational system, the ROC is outside the outermost pole. Since all poles are inside (|z|=1), the ROC includes the unit circle, so the system is BIBO stable."

The strong answer is better because it exposes the reasoning chain rather than only recalling a shortcut.

Likewise, for a pole-zero question:

Weak: "A zero causes a notch."

Acceptable: "A zero on the unit circle makes the response zero at that frequency."

Strong: “When (e^{j\omega}) reaches the angle of a zero on the unit circle, the distance to that zero is zero, so the numerator of (H(e^{j\omega})) becomes zero and a notch occurs.”

For a numerical exit ticket:

Weak: “3 kHz.”

Acceptable: “3 kHz because 9 kHz aliases when sampled at 12 kHz.”

Strong: “The Nyquist frequency is 6 kHz, so 9 kHz folds about 6 kHz to (12-9=3) kHz.”

A useful grading principle is therefore:

$$\boxed{\text{good exit-ticket answer} = \text{correct result} + \text{key justification}}$$

You generally do **not** need complete formal derivations, polished prose, or every intermediate algebraic step. In fact, if students need five minutes to write a paragraph, the question is probably doing more than an exit ticket should.

For your lectures, I would usually define a “good answer” as something that can be written in **one to three sentences or a few lines of math** and that demonstrates the central conceptual relationship you were trying to teach.